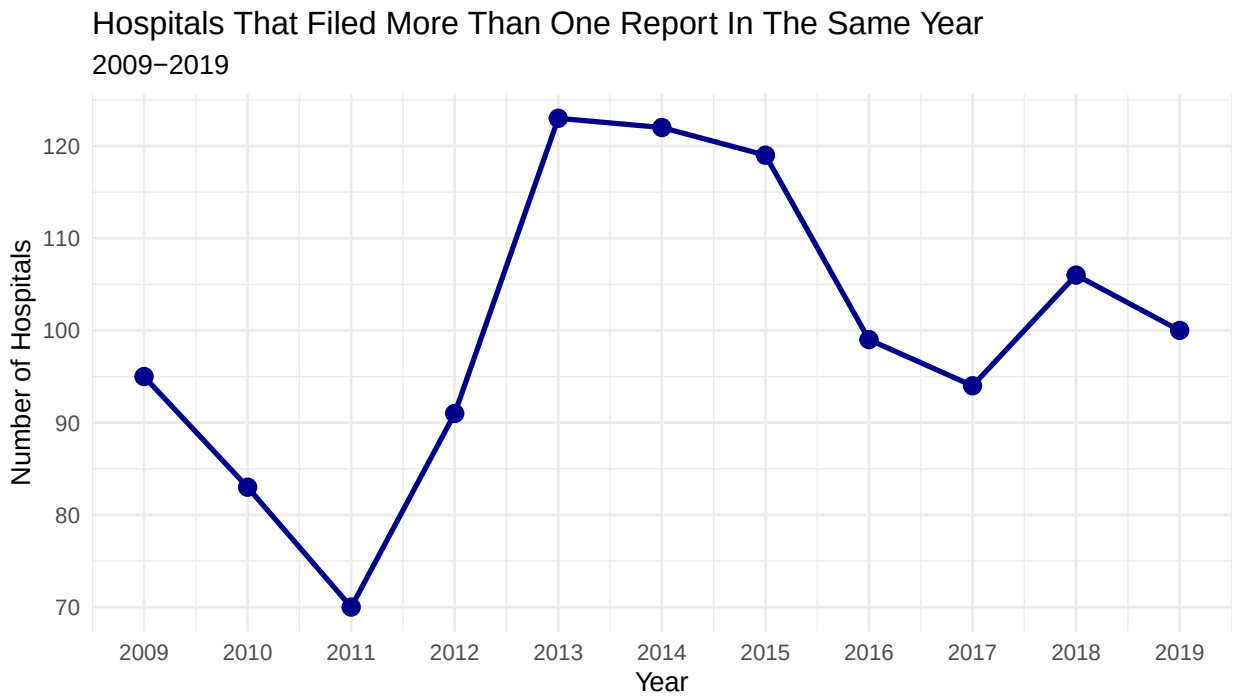
Homework 4: Submission 1

Mathew Kang

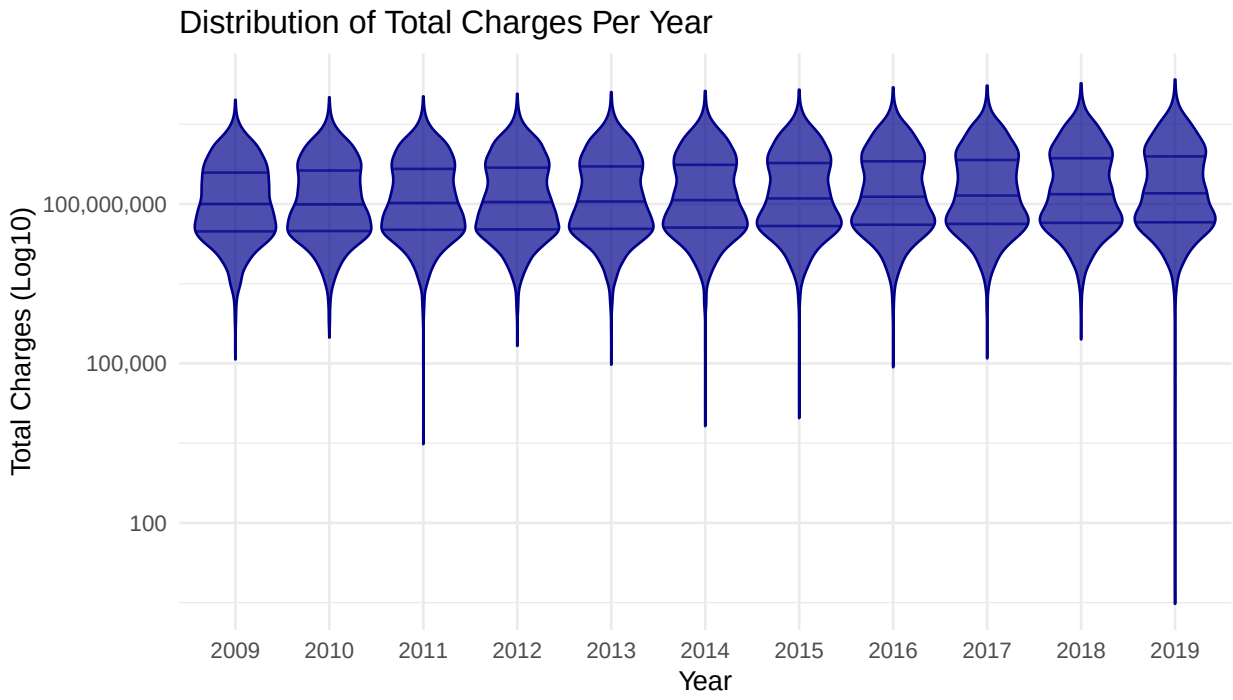
Question 1



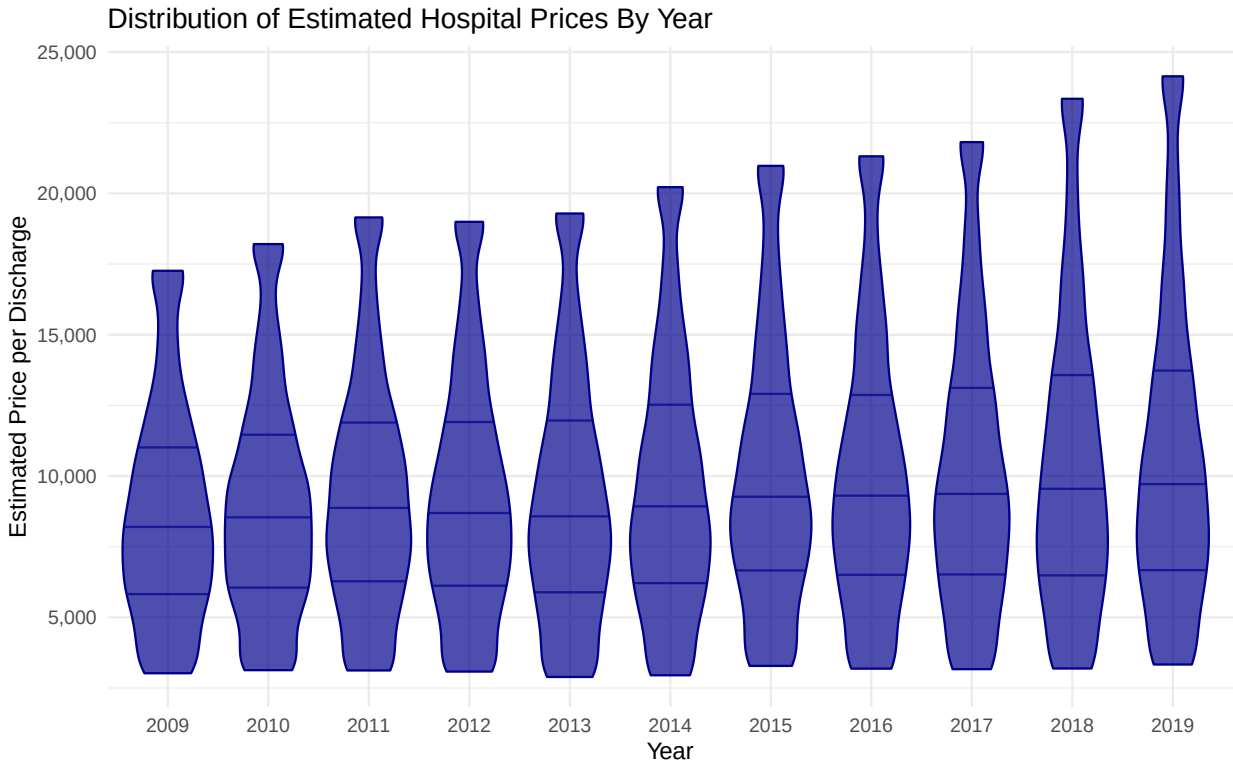
Question 2

Unique Hospitals
6874

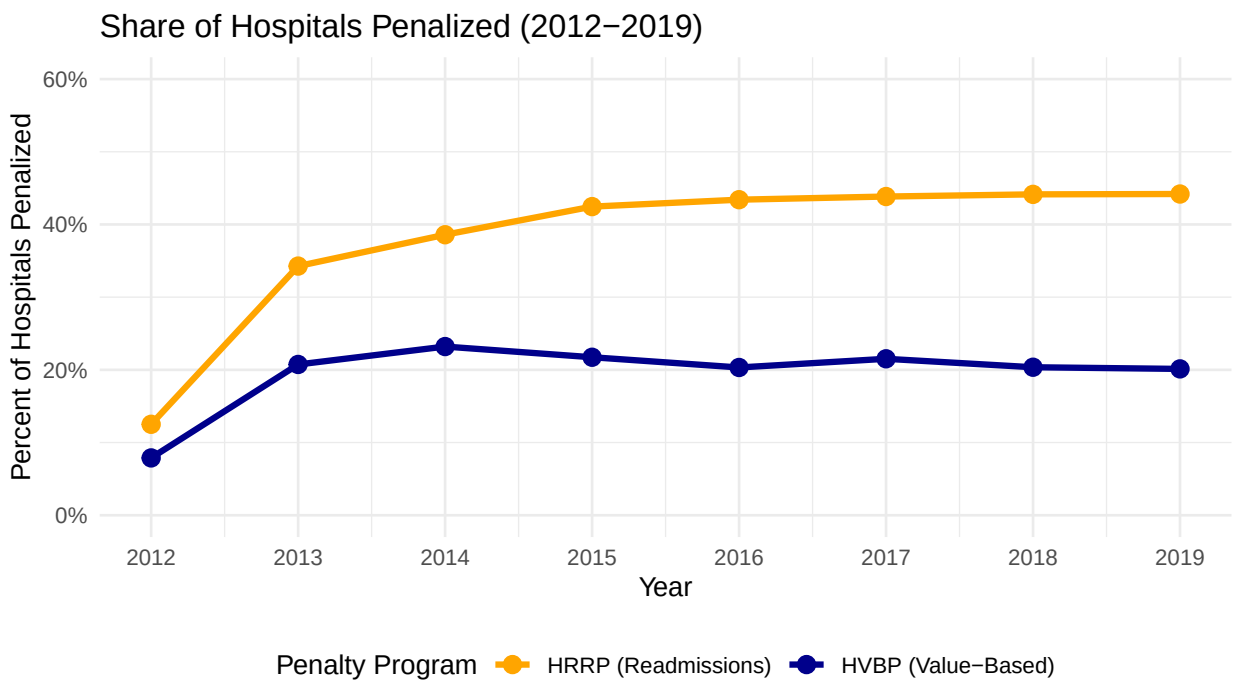
Question 3



Question 4



Question 5



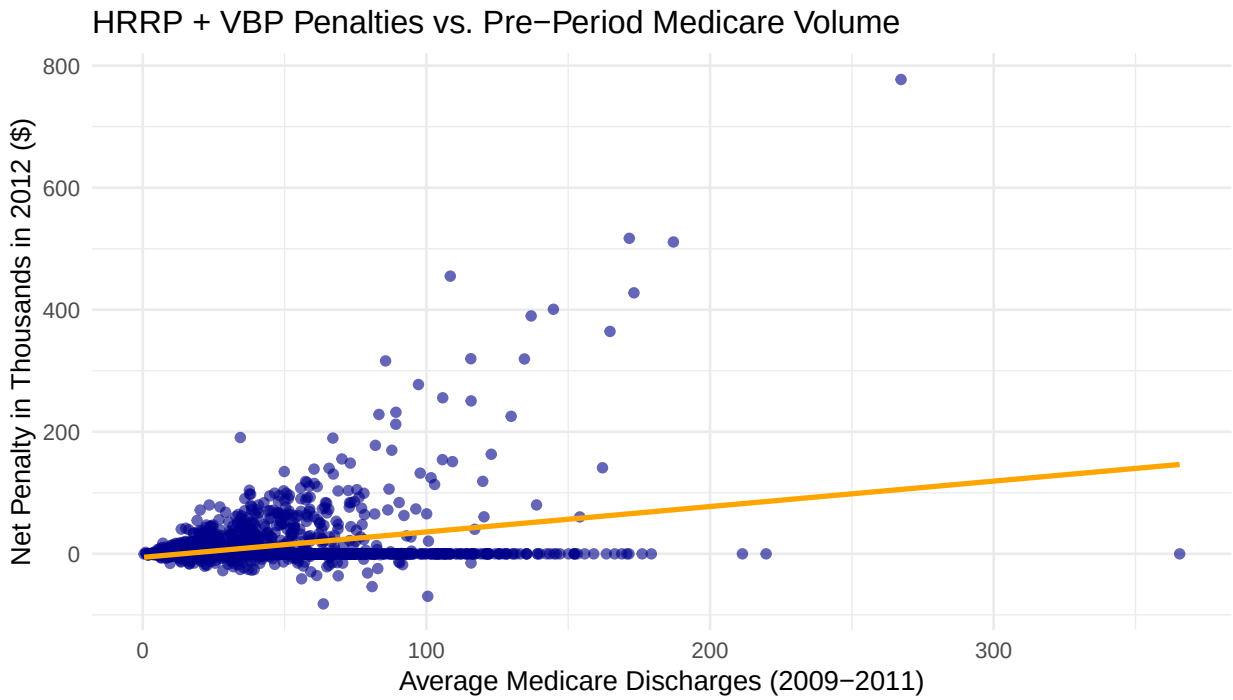
Question 6

Table 2: OLS Estimates of Net Penalty on Price Changes

	Baseline	+ Beds	+ Medicaid
Net Penalty ($1k)	5.075	4.872	2.410
	(13.175)	(13.185)	(9.529)
Avg. Beds		0.151	0.114
		(0.333)	(0.240)
Avg. Medicaid Discharges			0.048
			(0.176)
Num.Obs.	2466	2466	2455
R2	0.000	0.000	0.000

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Question 7



Question 8

Table 3: First Stage: 2012 Net Penalty on Pre-Penalty Medicare Volume

	Baseline	+ Beds	+ Medicaid
Avg. Medicare Discharges (per 100)	0.416***	0.417***	0.459***
	(0.024)	(0.024)	(0.027)
Avg. Beds		0.000	0.000
		(0.000)	(0.000)
Avg. Medicaid Discharges			−0.001**
			(0.000)
Num.Obs.	2466	2466	2455
R2	0.113	0.113	0.116

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 4: Reduced Form: Price Change (2011-2014) on Pre-Penalty Medicare Volume

	Baseline	+ Beds	+ Medicaid
Avg. Medicare Discharges (per 100)	37.294*	36.902*	30.535*
	(16.151)	(16.245)	(13.427)
Avg. Beds		0.076	0.074
		(0.332)	(0.238)
Avg. Medicaid Discharges			−0.166
			(0.199)
Num.Obs.	2501	2501	2490
R2	0.002	0.002	0.002

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Question 9

Table 5: 2SLS Estimates: Effect of Net Penalty on Price Changes (2011-2014)

	Baseline	+ Beds	+ Medicaid
Net Penalty ($1k)	89.783*	88.811*	66.849*
	(39.608)	(39.813)	(29.924)
Avg. Beds		0.078	0.073
		(0.337)	(0.243)
Avg. Medicaid Discharges			−0.087
			(0.187)
Num.Obs.	2466	2466	2455

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Question 10

When using prior Medicare discharges as the study instrument, the "local" ATE (LATE) measures the effect for "compliers," or hospitals whose penalty was determined by their pre-2012 Medicare volume, on the change in price per discharge (2011-2014). On the other hand, the ATE obtained from the OLS regression models measure the effect of the treatment, 2012 net penalties, on the aforementioned outcome. There are potentially two main drivers for the difference in estimates. First, the OLS regression understates the true effect by including wealthy, specialty hospitals that can easily absorb penalties without changing prices. Meanwhile, the LATE specifically capturing hospitals with large Medicare volumes implies larger estimates, since these hospitals do not have large cash reserves and must raise prices to continue operations. Second, the treatment is endogenous: penalties affect prices, and the opposite may be true when considering that large investments in quality improvement can reduce readmission rates.